



SOW
Project # 523-25-004
Replace Fire Alarm

1.0 Statement of Work

Provide all professional architectural and engineering (A/E) services necessary to develop a complete construction package, with investigations, specifications, drawings, cost estimates, project phasing, construction schedules, site visits and construction period services associated with continuation to replace all fire alarm devices in all buildings on campus. A/E to review existing FCA Report and provide corrections with final design submission.

1. The intents of this project are to include, but are not limited to:

1.1 Provide all professional architectural and engineering services necessary to Review for Bid completed drawings re-develop completed drawings, specifications, cost estimates, if required and project phasing associated with Replacement of the Fire Alarm System in all buildings on campus (All buildings total – 1,000,000M square feet). Patient Buildings have first priority, Administration/Staff Buildings have second priority, Utility and Support Buildings have third priority.

1.2 The designer services shall include, but not be limited to:

- Review of the existing campus network / hard wire system conditions.
- Creation of options related to systems design, peer-to-peer network, intuitive event driven UL 2572 mass notification standard etc.
- Design of new systems with including all panels, distributions, devices, smoke detectors, heat detectors, horns, strobes, pull stations, enunciators, signage, etc.
- Design of all support systems including electrical, mechanical, Historic Preservation Section 106 Compliance, and architectural elements shall be included.
- Through the course of the CODE AND CRITEREA REVIEW phase, it will be necessary to interface with Medical Center staff to ascertain the details of the scope through a series of meetings and periodic submissions of documents. All submission requirements shall conform to all VA criteria and standards, in addition to Commonwealth of Massachusetts state codes and regulations.
- Contractor to utilize 3D Lidar Scanning for above ceiling investigations. A/E to review FCA report and provide breakdown of corrective costs under this project.
- Construction ranging from \$5,000,000 and \$10,000,000.

1.3 Cost Estimating: The A/E is responsible, as part of his Quality Assurance Plan, to develop procedures for tracking estimated construction costs throughout the project design. The A/E shall notify the project engineer when estimated construction costs exceed the construction budget. The A/E shall be responsible for developing alternates to bring the project cost estimate under the construction budget. Cost estimates submitted as part of the required submittal requirements shall not be used as the sole method of meeting this requirement. The A/E shall modify project design as necessary to keep the project cost estimate at or below the construction funding level. These changes shall be made at no cost to the government.

1.4 Work will include, but not be limited to, demolition, new finishes, fire detection, and electrical

work.

1.5 Asbestos: The proper identification of Asbestos to determine construction costs associated with Comprehensive Strategy & Report phase of this project are considered part of this project.

1.6 To extent possible existing Asbestos Assessment of Building's will be made available to the A/E. Additionally, proper asbestos abatement design and Industrial Hygienist monitoring services required during construction to correct the deficiencies selected for the construction design portion of this project shall also be provided by the A/E.

1.7 Special attention will be required to develop a design that can be phased as to allow work to be performed while all services remain in operation and can reasonably maintain day to day operations. Through the course of the design phase, it will be necessary to interface with Medical Center staff to ascertain the details of the scope through a series of meetings and periodic submissions of documents. All submission requirements shall be in accordance with open ended A/E Contract and conform to all VA criteria and standards, in addition to Commonwealth.

2. Description of Work

2.1 General: Conduct preliminary meetings and site survey investigations to define the necessary elements of the project scope of work fulfilling the goals of the VA's project team (Engineering Service, VA Police, VA Safety), and presentation of preliminary layout plans with proposed alternatives to the User Group. Based on User Group meetings and site surveys, prepare a detailed project plan description report pertinent to the project requirements within the constraints of the construction budget consistent with the VA's project team goals. Through the course of the design phase, it will be necessary to interface with Medical Center staff to ascertain the details of the scope through a series of meetings and periodic submissions of documents. All design submission requirements shall be in accordance to Open ended A/E Contract and conform to the latest editions of all VA criteria/standards/regulations/design guides, etc. (VA's Office of Construction & Facilities Management web site: <http://www.cfm.va.gov>), in addition to Commonwealth of Massachusetts state codes and regulations and national codes and regulations.

2.2 Implementation: Engage the services of all consultants in each discipline deemed to be necessary to fulfill the requirements of the project plan. Each discipline shall document and investigate the relevant existing conditions found, identify/investigate/review the availability of existing building information, and field verify all essential elements pertinent to their discipline. Site investigations shall measure, photograph, video tape, and document the conditions of areas to be renovated and/or impacted. Provide the necessary investigation/inspection/testing/metering of existing utilities to ensure their adequacy for this project.

2.3 Project Development: The A/E shall provide all services required to prepare documents and reports for the subject project consistent with the VA's project team goals as prioritized during preliminary project meetings, field surveys and studies. The A/E shall determine existing conditions, severity of deficiencies, if any previous deficiencies have been corrected and to identify any conditions which may have previously been unreported. Cost estimates shall be prepared to reflect the cost of replacing the component in a stand-alone project (i.e., not as part of another project) and all cost for this replacement including but not limited to lead and asbestos removal, required phasing, special field conditions and/or temporary equipment needed during the construction. The completed report shall then be submitted for approval to the Project Engineer.

2.4 Project Design: Based on the approved design development the A/E shall provide all services required to prepare plans, specifications, and other supporting documents necessary to correct the identified deficiencies in priority order. The approved cost estimate and available project funding will be used to determine the amount of deficiencies which will be correct under this project. A complete design including but not limited to determination and evaluation of existing conditions, necessary engineering calculations, necessary design work, independent fire safety/life safety review, IH Services, quantity and cost estimates, bid schedule, construction cost estimates, a proposed construction schedule, design analyses and calculations, a design documentation report, engineering considerations and options report shall be developed for this project.

2.5 Asbestos & Lead: The proper identification Asbestos & Lead to determine construction costs associated with Comprehensive Strategy & Report phase of this project are considered part of this project. To extent possible existing Asbestos and Lead Assessment of the facility will be made available to the A/E. Additionally proper asbestos and lead abatement design and IH monitoring services required during construction to correct the deficiencies selected for the construction design portion of this project shall also be provided by the A/E.

3. General Description of Services —

A/E shall provide the services indicated in this Statement of Work, Supplement A, Supplement B and Supplement C. These services will be provided in distinct phases:

- Conceptual/15% Submittal – Preliminary User Group Meetings & Field Surveys
- 30% Submittal – Schematics
- 60% Submittal – “DD” Design Development
- 90% Submittal – “CD” Construction Documents
- 100% Submittal – Bid [Final] Documents Construction Period Services
- Project Close Out Submittal – “As-Built” Construction Documents

3.1 Drawings: The drawings, specifications, and all other submittal items will be prepared using English units of measure.

3.1.1 General: The A/E shall prepare drawings in a manner that clearly and adequately delineates the work to be accomplished by the construction contractor. Design documents will be sufficiently detailed to permit construction contractors to submit responsive bids without visiting the project site. The cover sheet will be signed and stamped by principal of the firm who is a registered Architect or Professional Engineer. All drawings will be created using AutoCAD and National CAD standards. These CAD standards are available on the Internet at the VA’s Office of Construction & Facilities Management web site: <http://www.cfm.va.gov> .

3.1.2 Electronic Files: One CAD (*.DWG) file shall be used per drawing (sheet) applying the VA’s CAD layering and disk labeling protocols. All files shall be IBM compatible and furnished on high quality CD-R disks (DWG + PDF per disk). Provide in accordance with Supplement “B” sets of drawings (AutoCAD DWG & Adobe Acrobat PDF). The computerized version, format and media of the computer disks and files shall be specified and verified by the VA Project Engineer at the time each submission is due (XRF, DXF and ZIP files are not acceptable). Provide durable high-quality plastic storage boxes at each submission for long term archiving of the disks.

3.1.3 Format: All drawings and sketches will be provided in both hard copy and electronic file format. Full-sized drawings shall be developed as “E” sized sheets (30 inches by 42 inches at the trim line) and shall utilize the standard VA title block. Fonts shall be standard AutoCAD, and drawings must be fully compatible with AutoCAD 2012. Drawing file names shall contain the drawing number and drawing title information. Drawings shall utilize world coordinate system with standard cross hairs. Each drawing shall have a separate drawing file name consistent with the drawing title. Adobe Acrobat PDF files for each AutoCAD file shall be provided.

3.1.4 Fire/Life Safety Review: The VA will conduct an independent fire safety and life safety review shall be accomplished at each design submission. Contractor to complete an independent review. Include the cost and coordination for a fire/safety review.

3.2 Specifications

3.2.1 General: The A/E shall utilize the latest edition of VA Master Specifications as revised and provided by the facility supplemented by the VA Office of Construction and Facilities Management Master Specifications as needed. The VA retains no responsibility for the development or preparation of

any specification. While it is recognized that certain Divisions and/or Sections will require significant input from the VA, the A/E retains overall responsibility for these Sections. The VA Project Engineer will provide common design comments to incorporate into the design that are specific to VA Boston Healthcare System.

3.2.2 Electronic Files: All files shall be IBM compatible and furnished on high quality CD-R disks. Provide in accordance with Supplement “B” sets of specifications (MS Word, PDF). The computerized version, format and media of the computer disks and files shall be specified and verified by the VA Project Engineer at the time each submission is due (ZIP files are not acceptable). Specifications shall be provided individually and as one compiled file both in MS Word and PDF for both. Provide durable high-quality plastic storage boxes at each submission for long term archiving of the disks.

3.2.3 Outline Specifications: The A/E shall develop an outline specification listing the proposed guide specifications and the A/E prepared sections that will be used for the project. The outline specification will list the guide specification number and title for each proposed section. Sections shall be arranged within their respective divisions, in numerical order. New specification sections, developed by the A/E, will be numbered to fall in their respective division at a logical location.

3.2.4 Editing Specifications: The A/E must thoroughly edit and adapt the Specifications to satisfy the project requirements and provide a complete set of construction specifications. The A/E shall notify the VA in writing should there be instances where there are no appropriate Master Specifications available for use. At the VA’s direction the required specifications will be prepared by the A/E. These specifications shall list the essential features, functions, and other factors to clearly indicate the type and quantity of items/work required. All specifications will be prepared by listing parameters and requirements that can be met by several manufacturers. The use of trade names may be used to establish the standard of quality to be met. To facilitate reviewing the following method shall be used until the Bid [Final] Documents are prepared ...

Deletions – Use strikethrough method

Insertions – Use double underline with blue text method.

3.2.5 Construction Contractor Submittal Register: The specifications require the construction contractor to submit shop drawings, samples, manufacturer’s data, certificates, test reports, and other items to the Government. The A/E shall prepare a complete listing of construction contractor submittal requirements on an Excel Spreadsheet. The spreadsheet shall have a separate line for each required submittal and separate columns for Division, Section, and type of submittal (catalog cut, sample, etc.). The register shall be incorporated with the specifications.

3.2.6. Specification Section Merging: In addition to providing individual specification sections (MS Word, PDF), final specification sections shall be merged into one merged specification document (MS Word, Adobe Acrobat PDF) files on high quality CD-R disks. Provide two (2) editable documents each with one (1) file containing full document pagination in the right-hand side of Footer without the loss of individual section Headers/Footers.

3.3 Quantity and Cost Estimates

3.3.1 Format: The A/E shall prepare quantity computations, cost estimates, and construction cost estimates for this project. All construction cost estimates shall conform to the requirements contained in VA Manual for Preparation of Cost Estimates (latest edition). Manual is available on the Internet at the VA’s Office of Construction & Facilities Management web site: <http://www.cfm.va.gov>.

3.3.2 Cost Estimate Submittal: Cost estimates shall be submitted in 2 hard copies and in electronic form (MS Excel, PDF). THE A/E IS RESPONSIBLE FOR ESTABLISHING INTERNAL CONTROLS FOR KEEPING THE PROJECT DESIGN WITHIN PROJECTED CONSTRUCTION FUNDING. THE A/E WILL BE REQUIRED TO REVISE DESIGN AND RESUBMIT COST ESTIMATES AS NECESSARY (AT NO COST TO THE VA) SHOULD COST ESTIMATES EXCEED CONSTRUCTION FUNDING.

3.4 Phasing and Construction Schedules

3.4.1 Schedules: The A/E shall prepare a proposed schedule for construction that is consistent with the project construction cost estimate. During development of this schedule, due consideration will be given to standard construction practices, durations of tasks, the sequence of construction, procurement of materials, climatic conditions, etc. It is planned that areas in and adjacent to the proposed construction area shall be occupied during construction. The A/E shall develop a phasing plan that addresses this fact and assures smooth implementation of required construction. Special attention will be required to develop a design that can be phased as to allow work to be performed while all services remain in operation and can reasonably maintain day to day operations.

3.4.2 Schedule Format: The construction schedules shall be horizontal bar charts that: a) identify the major construction activities required to perform the work; b) identify the duration of each of the major construction activities; c) identify the critical path network to achieve the work.

3.5 Design Analysis (DA): The A/E shall develop a DA that addresses general project parameters, functional and technical requirements, existing conditions, design objectives, design assumptions, product literature, and calculations applicable to the project's design. Design/Construction Alternatives, existing physical limitation and their impact shall also be included in the DA.

3.6 Quality Control

3.6.1. Quality Control Plan: The A/E shall prepare a Quality Control Plan (QCP) which includes the following as a minimum:

- Identification and discussion of all organizational and technical interfaces
- Design team members and their areas of responsibility
- Team members responsible for checking the design
- Team members responsible for checking the electronic files
- Team members responsible for reviewing and submitting the required VA Design Alerts and Quality Alerts
- Project Schedule showing key milestones and review periods

3.6.1.1. VA Design Alerts and VA Quality Alerts can be found on the internet at the VA's Office of Construction & Facilities Management web site: <http://www.cfm.va.gov>.

3.6.2. Quality Assurance: The A/E is responsible for developing and performing quality assurance reviews of all work to confirm that proper criteria, regulations, laws, codes, principles and professional procedures have been used. This includes work performed by subcontractors. The VA shall review all project submissions, but this review is general in nature and should not be considered as part of the A/E's Quality Control Plan and/or Quality Assurance.

3.6.3 Quality Certification: The A/E shall certify at each submission in writing that he/she has performed quality assurance reviews of that submission.

3.7 Site Visits, Meetings/Conferences, and Discussions

3.7.1. Site Visits: The A/E shall perform design site visits as required to accomplish the scope of this project. The purpose of these visits is to further define project requirements, observe and evaluate existing field conditions, review any existing documentation, conduct required design meetings, and to gather supplemental site data necessary for performing the design. Reports summarizing the conditions observed, personnel contacted, and data gathered during the visits shall be prepared and included in the Design Development Documentation. The A/E shall coordinate with the Project Engineer at least three (3) business days in advance with all the required information: summary of site visit, areas to be visited,

dates and time durations, access that will be required, personnel names and phone numbers, etc. so the Project Engineer can notify appropriate services.

3.7.2. Design Review Conferences: Design review conferences shall conduct shortly following the issuance of each review Submission. Refer to Supplement B for the proposed schedule but will typically take place two weeks after the submission date. Design review conferences for the Schematics, Design Development and Construction Document Development phase shall include the A/E Project Manager/Senior Designer. These conferences shall take place at the facilities where the work is being performed.

3.7.3 Site Visit and Meeting Minutes: The A/E shall take notes and prepare minutes for all site visits, meetings and conferences attended. Minutes will be prepared in typed form, signed by the A/E Project Manager, and furnished to project engineer within seven calendar days after the meeting/conference for concurrence and distribution to attendees. Copies of all meeting/conference minutes will be included in each Submission.

3.8 Responsibility after Design Completion: The A/E is required to support the VA should errors or omissions in the documents create problems in bidding or administering the contract for construction. As needed, the A/E will clarify the design intent and correct any errors or omissions in the original documents. The corrections shall be done in a timely manner at no additional cost to the Government. The A/E shall incorporate amendment changes on the original drawings and/or CAD drawings when requested to do so after the bidding process at no extra cost to the Government. In addition, the A/E shall incorporate amendment changes on the submittal registers and submit on a CD labeled with the project title, location, and construction contract number. Also, during the bidding period, the A/E is required to assist in answering all bidders' inquiries pertaining to the design. If clarifications are required, the A/E will prepare the required amendment. The A/E, however, shall not receive or respond to any direct inquiries from contractors.

3.9 A/E Services during Construction – Construction Period Administration

(CPA): CPA Services shall include, but are not limited to:

- review and approval of construction submittals covering products that have been listed within the contract documents and contractor substitutions, manufacturer's data/performance sheets, samples, shop drawings, schedules, phasing plans;
- provide and maintain an updated submittal register
- compatibility analyses of different materials and recommendations associated with acceptance or rejection of alternate materials and products;
- responses to Request for Information (RFI);
- change order review and recommendations;
- preparation of cost estimates to Request for Proposal (RFP);
- Cost/time/impact estimates and recommendations relevant to design omissions and construction plan deviations.
- site visits and Inspections as required (6 minimum) and/or requested by the VA Project Engineer for inspection of ongoing construction including final tests and inspections; documentation of all necessary site visits to the VA Project Engineer within one (1) week after the visit;
- verification of as-built conditions from contractor supplied marked up prints;
- preparation of as-built documents in accordance with Supplement "B" which the VA will retain permanently;
- IH services associated with any asbestos or lead abatement identified in the project.

The Government may issue a change order for the A/E to provide additional services, such as review of value engineering change proposals, preparation of design modifications, or other similar services during construction.

Electronic Construction Submittals:

The A/E shall have the capability to electronically review and process construction submittals including digitally signing the Material Approval Submittal. A PDF writer (such as Adobe Reader/Writer or Pro V Nuance, etc.) is required by the A/E in order to process electronic construction submittals.

The A/E shall provide and manage a file share site for the processing of electronic submittals. This file management site shall be set-up to allow the contractor to electronically submit the construction submittals, allow the A/E to review and process the submittals and allow the VA to review and process the submittals. Due to VA Security Protocols, some FTP and SharePoint sites are restricted. If this occurs, the A/E is responsible to establish and utilize another file share site. Central Desktop is one cloud-based project collaboration web-service that is currently supported by VA Security Protocols. ProjNet Dr Checks is also currently supported by VA Security Protocols.

4. References/Design Criteria

4.1 Incorporated References/Design Criteria — the following documents form a part of this contract to the extent referenced are included in the A/E Scope of Work. Design requirements identified in these documents are part of the A/E Required Services. National Codes, Standards and other requirements referenced in these documents also become part of the A/E Required Services. Design Copies of these may be found on the internet at the VA's Office of Construction & Facilities Management web site: <http://www.cfm.va.gov>.

Supplement "A" (attached)

Supplement "B" - A/E Submissions of Design Review Materials

(attached) Supplement "C" - A/E Minimum Submission Requirements

(attached) Asbestos Abatement Design Manual (*by referenced*)

ASHRAE Standard 55 – Thermal Environmental Conditions for Human Occupancy (*by reference*)

ASHRAE Standard 62 – Ventilation for Acceptable Indoor Air Quality (*by reference*)

ASHRAE Standard 90.1 – Energy Standard for Buildings Except Low-Rise Residential Buildings (*by reference*)

Barrier Free Design Guide (PG-18-13)

Construction Standards (H-18-3); latest revision (*by reference*)

Master Specifications (PG-18-1); as revised by facility supplemented with PG-18-1 as needed (*by reference*)

DVA National CAD Standard Application Guide (*by reference*)

DVA National CAD Standards & Detail (PG-18-4); latest revision (*by reference*)

Volume 1 Architectural; latest revision

Volume 2 Site Engineering & Landscaping; latest revision

Volume 3 Mechanical Engineering; latest revision

Section 1 Plumbing & Sanitary

Section 2 HVAC

Section 3 Steam Generation & Distribution

Volume 4 Electrical; latest revision

DVA Special Design Criteria; HVAC, Mechanical, Plumbing, Electrical, Structural, et al (*by reference*)

Environmental Planning Guidance (PG-18-7)

National Standard Plumbing Code (NAPHCC) (*by reference*)

National Electrical Code; latest revision (*by reference*)

Handicapped Standards; latest revision (*by reference*)

NFPA; latest revision (*by reference*)

NFPA 99, 101 and other applicable NFPA Codes; latest edition (*by reference*)

Applicable Occupational, Safety and Health Administration (OSHA) Standards (*by reference*)

Applicable VA Design Guides (*by reference*)

Applicable VA Design Alerts (*by reference*)

Applicable VA Design Manuals (*by reference*)
Applicable VA Quality Alerts (*by reference*)
Applicable VA Equipment Guide Lists (*by referenced*)
VA Equipment Reference Manual PG-18-6 (*by referenced*)
A/E Contract Procedures (*by reference*)
A/E Architectural Review Checklist (*by reference*)
A/E HVAC Review Checklist (*by reference*)
A/E Plumbing Review Checklist (*by reference*)
A/E Electrical Review Checklist (*by reference*)
A/E Coordination Drawing Checklist (*by reference*)
Manual for Preparation of Cost Estimates (*by reference*)
Applicable VA Energy Conservation & Sustainability Program requirements (*by reference*)
Preparation and Issuance of Construction Solicitation Contract Documents Manual (*by reference*)
A/E Submission Instructions for Minor and NRM Construction Program (PG-18-15) (*by reference*)
Physical Security Design Manual
Room Finishes, Door, and Hardware Schedules (PG-18-14)
Site Utility Design Manual

4.2 Energy Requirements for all New Construction, Major & Minor Renovation Projects (ASHRAE Standard 90.1-2007 Energy Standard for Buildings):

4.2.1 Major and minor Construction:

All new major & minor construction projects shall be designed to achieve an energy consumption level that is at least 30% below the level achieved under ASHRAE Standard 90.1-2007, if life cycle cost effective. Use Appendix G of ASHRAE Standard 90.1-2007 to document the energy savings.

Should the 30% reduction in energy consumption prove not to be life cycle cost effective, then the AE shall evaluate alternative designs at 5% successive decrements in order to identify the most energy efficient design that is life cycle cost effective.

The following parameters shall be used for performing the analysis:

- 20-year life-cycle period for system comparisons
- Use Public domain programs such as TRACE, E-CUBE, Carrier E20-II, Energy Plus, etc.
- 3% Discount Factor; and
- Neither method shall include taxes or insurance costs when computing cost.

4.2.2. Major and minor renovations and Major Equipment Replacements:

For major renovations: Shall include any renovation where an entire area is "gutted" and rebuilt, regardless of funding source (Major, Minor, NRM or CSI). Any replacement of central HVAC system or changes in air distribution shall be considered a "major Renovation."

Reduce the energy cost budget by 20% below pre-renovations 2003 baseline. AE shall estimate 2003 baseline energy consumption before the renovation and compare it to the proposed design after the renovation and document the 20% savings. The primary unit for energy budget reporting is the British Thermal Unit (BTU) per square foot of the area to be renovated. The AE shall develop the energy budget as required in ASHRAE 90.1-2004, Appendix G, using the same parameters outlined above for new construction.

For NRM and CSI Projects: Where a small portion of a building, and only architectural treatments, or simple equipment replacement is planned. The renovation will have little to no impact on the central HVAC System, or energy use within the renovated area. The AE shall design all renovated areas to comply with ASHRAE Standard 90.1 -2007 as a minimum. Design to better than ASHRAE Standard 90.1 -2007,

when obviously life cycle cost effective. Appendix G analysis is not required for NRM and CSI projects.

Any part of the renovation which impacts energy consumption shall be designed with the mandate to reduce energy consumption. For example, if exterior walls are part of the project, then their R- Value shall be increased. If lighting fixtures are to be replaced, then they shall be replaced with the most energy efficient fixture that is cost effective (20-year life) and meets the lighting W /SF required in Chapter 9, ASHRAE Standard 90.1 -2007.

Equipment shall meet the Mass Save and NGrid prescriptive rebate requirements. Contractor to submit completed application with equipment submittals that AE shall review. The VA is to receive the rebate as a utility bill credit.

4.2.3 Renewable Energy Resources: For any work, renewable (geothermal, heat recovery, wind, PV solar, solar hot water, etc.) shall be evaluated for life-cycle cost effectiveness.

5. Design Budget

5.1 A/E is responsible to develop a design that can be built within budget and function as intended without change orders. The design must be based on the reality of existing site conditions and not solely on "As-Built" drawings. The A/E as part of their site investigation is responsible for confirming all existing conditions that may affect the project construction.

5.2 Cost Estimating: A/E shall be responsible, as part of the Quality Assurance Plan, to develop procedures for tracking estimated construction costs throughout the project design. Cost estimates submitted as part of the submission requirements shall not be used as the sole method of meeting this requirement. The A/E shall notify the Project Engineer when estimated construction costs exceed the construction budget. The A/E shall be responsible for proposing alternates to bring the project cost estimate within the construction budget. A/E shall modify design elements as necessary to keep the project cost estimate within the construction funding level. These changes shall be made at no additional cost to the Government.

5.3 Construction Budget: \$7,983,574.00

5.4 Deduct Alternates: Shall be utilized in the best interest of the Government for market condition variations to maximize the use of available Government funds. The Government shall select the proposed alternates from the A/E recommendations. Deduct Alternates shall not be used to bring the projected project costs within construction funding. These alternates are designed to assure a contract award and are in addition to the requirement to have a project design and estimate at or below construction funding. The A/E shall incorporate deduct alternates totaling at least 20% of the construction estimate into the project. The deduct alternates shall be clearly identified in the specifications General Requirements section and on the drawings.

5.5 Add Alternates: Shall not be used.

6. Design Phases

6.1 Conceptual with Alternatives: Refer to Supplement A and B. Working with the VA Medical Center's project team (Project Engineer and User Group) the A/E shall propose conceptual needs and requirements with proposed alternatives for the planned scope of work. This task is an iterative process requiring multiple proposals and revisions developed during site meetings.

6.2 Schematics: Refer to Supplements A, B and C. A/E shall develop specific design needs and requirements with proposed alternatives. These documents are to be developed from the documents, meeting minutes and recommendations made during the conceptual phase. Any existing utilities and/or equipment which are to be incorporated into the project design shall be reviewed and evaluated to assure

that they are indeed there and are in sufficient quantity and condition to do so. Documents for this task define project requirements based on the planned scope of work. All required functional requirements shall be identified and incorporated into the project. All Architectural and Engineering requirements shall be identified and incorporated into the project. A/E shall provide an evaluation of the coordination of the drawings between disciplines; identification of errors, omissions or conflicts found within or between drawings and specifications.

6.3 Design Development: Refer to Supplements A, B, C. Any existing utilities and/or equipment which are to be incorporated into the project design shall be further reviewed and evaluated to assure that they are indeed there and are in sufficient quantity and condition to do so. Documents for this task define project requirements based on the approved scope of work. These documents are to be developed from the documents, meeting minutes and recommendations made during the schematics phase. All required functional requirements shall be identified and incorporated into the project. All Architectural and Engineering requirements shall be identified and incorporated into the project. A/E shall provide an evaluation of the coordination of the drawings between disciplines; identification of errors, omissions or conflicts found within or between drawings and specifications. A/E shall provide an evaluation of schedule to meet operating plan, project phasing and market conditions which may affect bidding. It is expected that at the completion of this phase marks the completion of the data gathering portion of the project. The A/E shall have obtained all information required (including site visits) to proceed directly the preparation of Construction Documents.

6.4 Construction Documents: Refer to Supplement C. The A/E shall develop complete drawings and specifications and other documents necessary for the bidding and construction of this project. These documents are to be developed from the documents, meeting minutes and recommendations made during the design development phase. A/E shall perform an evaluation of the coordination of the drawings between disciplines; identification of errors, omissions or conflicts found within or between drawings and specifications. A/E shall provide an evaluation of schedule to meet operating plan Invitation for Bids and Award dates, project phasing, and market conditions which may affect bidding.

6.5 Special Remarks: Cost estimates shall incorporate construction related issues which include but are not limited to materials, systems and construction techniques. Design documents review, analyses and recommendation of cost estimates covering labor, materials, equipment, general conditions and requirements developed for the project. Estimates shall allow for incorporation of review comments, which are within the Statement of Work.

7. Submittals and Performance Schedule

7.1 Deliver Submission Documents to the VA Boston Healthcare Project Engineer (MS-138P). The performance period will be project specific and will be specified with each scope of work.

7.2 Submission Materials and Schedules: Refer to Supplement B. Reproduction, mailing and delivery time is inclusive to the stated durations.

7.3 Government Review, Comment Resolution, and Meeting Minutes: The VA will review all submittals identified under this contract for functional and aesthetic relationship. The VA review may consist of written and/or oral comments at the discretion of the VA. VA review of the A/E's work shall not be construed by either party to relieve the A/E from any requirements of this scope of work; the A/E is responsible for all design work. The A/E shall submit written responses to the VA's oral and written design review comments; these written responses shall be within seven (7) calendar days of the VA's comments; and these responses shall explain the action the A/E will take for each comment. The A/E will be responsible for taking minutes at meetings and submitting them to the project engineer for concurrence within seven (7) calendar days of the meeting. Both parties will discuss these comments, if necessary, and attempt to resolve any unsettled issues that may arise from the review. Time for reproduction and mailing is inclusive to the stated durations.

7.4 The A/E may choose to perform work, at its own risk, during the Government review and comment resolution period, however, comments resulting from that review must be incorporated into the design prior to the next submittal. In the event a subsequent design phase is not authorized, the A/E shall incorporate all available review comments into the design to complete the current phase.

7.5 Construction Execution:

Construction Documents to Bidders Date: To Be Determined (TBD)

Bid Opening Date: TBD

Actual Construction Contract Award: TBD

Anticipated Construction Duration: TBD

Construction Completion Date: TBD

8. Authorized Changes — The A/E shall only accept instructions and/or directions from the assigned Project Engineer. Changes in the project scope of work must be authorized by the VA Contracting Officer.

9. A/E Data Use Responsibilities - Due to the scope of work and the exchanging of VA security data subject to restrictions, the VHA BHS Privacy Officer and Information Security Officer require the AE to follow the VA Handbook 6500.6 Appendix A, which includes the following item(s):

Data Use Agreement (DUA) – an agreement document used for the transfer of data that has been developed by the government, where the data is nonpublic or is otherwise subject to some restrictions on its use. This will be established in union with the VHA BHS Privacy Officer and Information Security Officer.

Business Associate Agreement (BAA) - an agreement that defines the contractor's responsibility as to government data. This will be established in union with the VHA BHS Privacy Officer and Information Security Officer.

Proper A/E Training Certificates to be in place. This will be coordinated with the VHA BHS Privacy Officer and Information Security Officer.

Security Background checks to occur. This will be coordinated with the VHA BHS Privacy Officer and Information Security Officer.

SUPPLEMENT “A”

Additional Requirements of Architect/Engineer Services:

- 1) On site visits to define the details of the project scope of work meeting the goals of the Veterans Affairs' (VA) project team, and presentation of preliminary layout plans with proposed alternatives.
- 2) Electronic drawings shall be developed utilizing most up to date VA version of AutoCAD. Each electronic drawing file shall be identified with the drawing number and drawing title. Electronic submission shall include AutoCAD DWG & Adobe Acrobat PDF per drawing (sheet).
- 3) Specifications shall be developed utilizing most up to date VA version of MS Word. Electronic Specification files shall be identified with the specification number and specification section title. An individual MS Word specification section file as well as a consolidated MS Word file

of all specification sections is required. In addition, a PDF consolidated file of all specifications sections is required. Provide two (2) editable documents each with one (1) file containing full document pagination in the right-hand side of Footer without the loss of individual section Headers/Footers.

- 4) Cost Estimates shall be developed utilizing most up to date VA version of MS Excel.
- 5) All computer-generated files shall be IBM compatible and furnished on high quality CD-R disks. Provide in accordance with Supplement “B” sets of drawings (AutoCAD DWG & Adobe Acrobat PDF) and specifications (MS Word and PDF) and cost estimates (MS Excel, PDF). The computerized version, format and media of the computer disks and files shall be specified and verified by the VA Project Engineer at the time each submission is due (XRF and ZIP files are not acceptable). Provide durable high-quality plastic storage boxes at each submission for long term archiving of the disks by the VA.
- 6) 90% Design Submission: The 90% design submission shall be a **Complete 100% Design**. All plans, sections, details, etc. shall be complete at this time. The A/E shall ensure that the 90% design submittal is in fact his/her 100% product, and that it represents a total biddable package. An incomplete 90% design submittal may be rejected by the VA; a rejection will require the A/E to resubmit a 90% design with no additional cost to the government.
- 7) The design consultant must have a PDF writer program to electronically sign material submittals; all Material Submittal Forms will be processed via electronic format with digital signature which utilizes a PDF Writer Program such as Adobe Reader/Writer or Pro V Nuance.
- 8) Site visits (8 for design and 10 for construction) as required and/or requested by the VA Contracting Officer and Project Engineer for review and inspection of progress including final tests and inspections.
- 9) Phasing and Construction Project Schedules shall be developed using most up to date VA version of Microsoft Office Project or Microsoft Office Excel.

SUPPLEMENT “B”

A/E SUBMISSIONS OF DESIGN REVIEW MATERIAL

FIRST REVIEW (Conceptual – Preliminary User Group Meetings & Field Surveys) Hardcopy:

2 Sets of reduced size preliminary layout plans with proposed alternates per meeting - 15 inches by 22 inches

- 2 Sets of bound conceptual estimates with proposed alternates per meeting
- 2 Sets of preliminary investigation report with recommendations and alternatives per meeting
- Electronic: 2 Sets of high-quality CD-R disks with the following on each:
 - 1 Set of PDF preliminary layout plans with proposed alternates per meeting
 - 1 Set of PDF conceptual estimates with proposed alternates per meeting
 - 1 Set of preliminary investigation report with recommendations and alternatives

SECOND REVIEW (30% Submittal – Schematics)

Hardcopy:

- 1 Set of full size bound schematic layout plans/details – 24 inches by 36 inches
- 1 Set of reduced size bound schematic layout plans/details on 20# bond
- 1 Set of 3-ring bound draft specifications with Mylar front and back cover sheets
- 1 Set of bound schematic estimates
- 1 Set of draft phasing plans

Electronic:

- 2 Sets of high-quality CD-R disks with the following on each:
 - 1 Set of AutoCAD & PDF schematic drawings in accordance with Supplement “A”
 - 1 Set of MS Word & PDF draft specifications in accordance with Supplement “A”
 - 1 Set of MS Excel & PDF schematic estimates in accordance with Supplement “A”

1 Set of Design Analysis with investigation reports, recommendations, alternatives and calculations

THIRD REVIEW (60% Submittal – “DD” Design Development)

Hardcopy:

- 2 Sets of full-size bound design development layout plans/details – 24 inches by 36 inches
- 1 Set of reduced size bound design development layout plans/details on 20# bond
- 1 Set of 3-ring bound design specifications with Mylar front and back cover sheets
- 1 Set of bound design development estimates
- 1 Set of design development phasing plans

Electronic:

- 2 Sets of high-quality CD-R disks with the following on each:
 - 1 Set of AutoCAD & PDF design drawings in accordance with Supplement “A”
 - 1 Set of MS Word & PDF design specifications in accordance with Supplement “A”
 - 1 Set of MS Excel & PDF design development estimates in accordance with Supplement “A”
 - 1 Set of Design Analysis with investigation reports, recommendations, alternatives and calculations

FOURTH REVIEW (90% Submittal – “CD” Construction Documents)

Hardcopy:

- 1 Set of full-size bound construction drawings layout plans/details – 24 inches by 36 inches
- 1 Set of reduced size bound construction drawings layout plans/details on 20# bond
- 1 Set of 3-ring bound construction specifications with Mylar front and back cover sheets
- 1 Set of bound construction estimates
- 1 Set of construction documents phasing plans

Electronic:

- 2 Sets of high-quality CD-R disks with the following on each:
 - 1 Set of AutoCAD & PDF construction drawings in accordance with Supplement “A”
 - 1 Set of MS Word & PDF construction specifications in accordance with Supplement “A”
 - 1 Set of MS Excel & PDF design development estimates in accordance with Supplement “A”
 - 1 Set of Design Analysis with investigation reports, recommendations, alternatives and calculations

FIFTH REVIEW (100% Submittal – Bid [Final] Documents)

Hardcopy:

- 1 Set of full size unbound final bid plans/details on Mylar with Registered Architect (RA) and Professional Engineer (PE) stamps - 30 inches by 42 inches
- 1 Set of full size bound final bid plans/details on 20# bond with RA & PE stamps
- 1 Set of reduced size bound final bid layout plans/details on 20# bond with RA & PE stamps
- 1 Set of 3-ring bound final specifications paginated front to back with Mylar front and back cover sheets
- 1 Set of bound final estimates
- 1 Set of final phasing plans
- 1 Set of final Design Analysis with investigation reports, recommendations, alternatives and calculations

Electronic: 2 Sets of high-quality CD-R disks with the following on each:

- 1 Set of AutoCAD & PDF final bid drawings in accordance with Supplement “A”
- 1 Set of MS Word & PDF final specification Sections in accordance with Supplement “A”
- 1 Set of MS Word & PDF final specification Sections merged in accordance with Supplement “A”
- 1 Set of MS Excel & PDF final estimates in accordance with Supplement “A”
- 1 Set of final Design Analysis with investigation reports, recommendations, alternatives and calculations

SIXTH REVIEW (Project Close Out Submittal – “As-Built” Construction Documents)

Hardcopy:

- 1 Set of full size As-Built plans/details on Mylar with RA & PE stamps
- 1 Set of reduced size bound As-Built plans/details on 20# bond with RA & PE stamps

Electronic:

- 2 Sets of high-quality CD-R disks with the following on each:
 - 1 Set of AutoCAD & PDF As-built drawing in accordance with Supplement “A”

Project Design Schedule

Facility: DVA – VA Boston Healthcare System, Jamaica Plain Campus

A/E Firm: TBD

Project Title: Replace Fire Alarm

Project Number: 523-09-152

Duration: 130 Calendar Days after Notice to Proceed (NTP)

ID	Milestone Task Description	Duration	Schedule Date	Actual Date
1	User Group Preliminary Mtg (Review Scope, Site Survey Investigation)	1 Day	NTP + 1	
2	Preliminary Investigation of Existing Conditions	As Needed		
3	Submit Conceptual Layout Schemes with Proposed Alternates	---		
4	User Group Mtg (Review Proposed Schemes)	---	NTP + 11	
5	Submit Revised Conceptual Layout Plans with Proposed Alternates	---		
6	User Group Mtg (Review Conceptual Plans)	---	NTP + 25	
7	Submit Conceptual Layout Design for Approval	---		
8	User Group Mtg (Review/Approve Conceptual Plans)	1 Day	NTP + 35	
9	Prepare 30% Schematic Design Submissions	As Needed	NTP + 50	
	Survey Existing Conditions, Coordinate Design with Consultants & VA Comments			
10	Submit 30% Schematic Design Package	---		
	VA Engineering 30% Review	14 Days (Min)		
11	User Group 30% Schematic Design Review Meeting	1 Day	NTP + 65	
12	Prepare 60% "DD" Design Development Submissions	---		
	Confirm Existing Conditions, Coordinate Design with Consultants & VA Comments	As Needed		
13	Submit 60% "DD" Design Development Package	---		
	VA Engineering 60% Review	14 Days (Min)		
14	User Group 60% "DD" Review Meeting	1 Day	NTP +85	
15	Prepare 90% "CD" Construction Document Submissions	---		
	Coordinate Design with Consultants, Existing Conditions & VA Comments	As Needed		
16	Submit 90% "CD" Construction Documents Package (A/E 100% design)	---		
	VA Engineering 90% Review	14 Days (Min)		
17	User Group 90% "CD" Review Meeting	1 Day	NTP + 105	
18	Prepare 100% Bid Final Documents	---		
	Finalize Coordination with Consultants, Existing Conditions & VA Comments	As Needed		
19	Submit 100% Bid Final Documents with RA and PE Stamps	---	NTP + 130	

SUPPLEMENT "C"

A/E Minimum Submission Requirements

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DISTRIBUTION OF A/E MATERIAL

**PROGRAM
MEDICAL CENTER PROJECTS**

I. GENERAL

A. INTRODUCTION

1. This document contains information and minimal submission requirements for contract documents specified in the A/E contract.
2. The Department of Veterans Affairs (VA) may contract with an Architect/Engineer (A/E) for any portion of a design: Schematics, Design Development, Construction Documents, or a combination of these.
 - a. For projects where the VA is contracting for Schematics Documents only, Schematics and Design Development Documents only, or Schematics, Design Development, and Construction Documents, the VA will provide the Design Program (if available), Facility Development Plans (if available), and VA design standards to accompany the Scope of Work for the project.
 - b. For projects where the VA is contracting for Design Development and Construction Documents only, the VA will provide the Schematics Plans and VA design standards to accompany the Scope of Work for the project.
 - c. For projects where the VA is contracting for Construction Documents only, the VA will provide the Design Development Plans and VA design standards to accompany the Scope of Work for the project.
3. Coordinate all activities with the VA Medical Center (VAMC). Hold informal meetings (upon mutual consent of the VA and the A/E) at the VAMC to discuss the design and related issues. Continue to expand contacts by telephone, rough sketch studies and other means of communication with the purpose of finalizing a general design approach to be followed.
4. Final approved Schematic documents shall be the basis for the development of the Design Development phase. Likewise, final approved Design Development documents shall be the basis for the development of the Construction Documents phase. The VAMC must approve any changes from each set of documents before the A/E proceeds to the next phase.
5. VA will review all submittals for functional and aesthetic relationships. However, no further functional decisions are anticipated after the Design Development phase.
6. Provide a design narrative/analysis for each technical discipline (e.g., architectural, mechanical, fire protection, etc.) which describes the intent of each discipline with schematic and/or design development submission.

7. Provide computations and sizing calculations for electrical, mechanical (HVAC, plumbing, and steam), sanitary, structural and fire protection designs. For computerized calculations, submit complete and clear documentation of computer programs, interpretation of input/output, and description of program procedures.
8. Provide individually packaged drawings for each submission to each unit specified in the "Distribution of A/E Materials" section.
9. Submit a complete set of final approved drawings incorporating all revisions, as per Supplement B A/E Submissions of Design Review Material after completion of the Schematics and Design Development stages.
10. At each review stage, the VA's technical reviewer, a value-engineering consultant, or a construction manager will perform a value engineering review.
11. Submit final drawings (Bid Documents) in high quality CD-R disks to be used with the AutoCAD version at the VAMC. Submit instructions on the use of the disks along with a complete listing of all layers that are used.

B. A/E RESPONSIBILITIES:

1. Contract documents shall meet or exceed the requirements of this document.
2. The A/E is responsible for producing a complete set of drawings, design narrative/analysis, calculations, sample boards, and specifications in accordance with professional standard practices and VA criteria. Each A/E discipline shall receive a copy of their respective VA design manuals, standard details, construction standards, and VA National CAD Standard Application Guide. The AE is responsible for obtaining the NCS.
3. A/E shall conduct coordination meetings between A/E technical disciplines before submitting material for each VA review and provide minutes of the meetings to VAMC.
4. A/E shall adhere to the approved Memorandum of Agreement (MOA).
5. A/E shall provide a checklist of all submittals, certifications, tests, and inspections required per drawing and specification section.
6. In addition, the A/E shall conduct interim fire protection installation inspections and witness final fire protection equipment testing.
7. AE Data Use Responsibilities - Due to the scope of work and the exchanging of VA security data subject to restrictions, the VHA BHS Privacy Officer and Information Security Officer require the AE to follow the VA Handbook 6500.6 Appendix A, which includes the following item(s):
 - Data Use Agreement (DUA) – an agreement document used for the transfer of data that has been developed by the government, where the data is nonpublic or is otherwise subject to some restrictions on its use. This will be established in union with the VHA BHS Privacy Officer and Information Security Officer.
 - Business Associate Agreement (BAA) - an agreement that defines the contractor's responsibility as to government data. This will be established in union with the VHA BHS Privacy Officer and Information Security Officer.
 - Proper AE Training Certificates to be in place. This will be coordinated with the VHA BHS Privacy Officer and Information Security Officer.
 - Security Background checks to occur. This will be coordinated with the VHA BHS

Privacy Officer and Information Security Officer.

C. SUBMISSION POLICY:

1. There is a Schematic* submission, a Design Development (DD**) submission, and a Construction Document (CD***) submission indicated in this guide. The VAMC may alter the submission requirements depending upon the complexity of the project by adding or deleting certain reviews. Where additional reviews might be required, the VAMC will issue, at their discretion, a detailed “Statement of Task” or supplemental instructions to the A/E, which would be provided at the time of solicitation for a fee proposal.
2. At each submission, the A/E shall date all material and present the designs on VA standard size drawings that are appropriately labeled, "SCHEMATIC SUBMISSION", "DESIGN DEVELOPMENT SUBMISSION", OR "CONSTRUCTION DOCUMENT SUBMISSION", in large block letters above or beside the VA standard drawing title block. In each submission, the A/E shall incorporate the corrections, adjustments, and changes made by VA at the previous review.

D. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC):

In an effort to reduce construction change orders due to design errors and omissions, the Office of Facilities Management has initiated a Quality Assurance/Quality Control program. The A/E shall develop, execute, and demonstrate that the project plans and specifications have gone through a rigorous review and coordination effort. The requirements are as follows:

1. Fee Proposal: Provide an outline of the actions that your firm will take during the design process along with an associated fee.
2. Two Weeks after Receipt of the Notice to Proceed: Submit a detailed QA/QC Plan describing each step that will be taken during the development of the various phases of design. Each step should have an appropriate space where a senior member of the firm can initial and date when the action has been completed.
3. 100% Submittal: Submit the completed QA/QC Plan along with the latest marked-up documents (plans, specifications, etc.) necessary to ensure that a thorough review and coordination have been completed.

E. ADDITIONAL SERVICES:

If additional services (i.e. surveys, soil borings, asbestos surveys, water flow testing, or lead surveys), are necessary to be performed by consultants, submit criteria for the work to be performed to the VAMC Contracting Officer as soon as possible. Upon approval of the criteria, submit proposals and qualifications of at least three firms being considered for the work in accordance with the contract procedures (CP1) of the contract, together with a proposal from the recommended firm and a brief justification for its selection, for VA approval. A/E should submit survey information for the Schematic Review.

F. CRITICAL PATH METHOD PHASING MEETINGS

A. If required and prior to submission of Schematic material, the A/E shall meet with the VAMC's Project Manager to discuss and outline phasing requirements for the project. These phasing requirements shall describe the general sequence of the project work, estimated project duration, and what Government constraints will exist that will influence the Contractor's approach to the construction project. The A/E shall be responsible for recording the phasing requirements.

B. Submit a phasing narrative and phasing plans (on reduced size plans) within two weeks after each phasing meeting to the VAMC Project Manager. VA will review these submission(s) and return comments to the A/E within two weeks of receipt. The A/E will then use this information in preparing their schematic, design development, and construction document submissions.

II. SUBMISSIONS

A. **SITE DEVELOPMENT:** Submit the following as applicable for the project:

Site Development:	Schematics*	DD* *	CD** *
Design Analysis (Narrative)	ü	ü	ü
Analysis of site	ü		
Circulation study	ü		
Phasing analysis	ü		
Parking analysis	ü		
Development concept showing proposed buildings and structures	ü		
Landscape drawings with plant groupings	ü		
Topographic, utility, and landscape survey		ü	ü
Demolition plan	ü	ü	ü
Layout plan showing location of:			
• Building and structures	ü	ü	ü
• Roads	ü	ü	ü
• Fire Access		ü	ü

Site Development:	Schematics*	DD* *	CD** *
• Parking	ü	ü	ü
• Accessible spaces		ü	ü
• Van spaces		ü	ü
• Mechanical and electrical equipment on grade	ü	ü	ü
• Future expansion	ü		
• Off-site roads	ü	ü	ü
• Off-site utilities	ü	ü	ü
• Service area(s)		ü	ü
• Entrances and exits		ü	ü
• Walks		ü	ü
• Inlets		ü	ü
• Contractor's staging area		ü	ü
• Vertical and horizontal road alignment		ü	ü
• Paving joint patterns		ü	ü
Grading plan showing:			
• Existing contours		ü	ü
• Proposed contours		ü	ü
• Spot elevations at structure corners, entrances, equipment pads, etc.		ü	ü
• First floor elevations		ü	ü
• Rim and invert elevations on storm drainage fixtures		ü	ü
• Erosion and sediment control		ü	ü
Rock excavation (quantity)		ü	ü
Planting plan showing:			
• List of plant material	ü	ü	ü
• Limits of irrigation	ü	ü	ü
Site details		ü	ü
Landscape details		ü	ü
Signage plan and schedule		ü	ü
List of Required Specifications	ü		
Specifications		ü	ü
Component catalog cuts	ü	ü	ü

* Submit site and landscape plans at an appropriate scale to show all work involved.

** Submit site and landscape plans at same scale as topographic/utility survey incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated site and landscape plans incorporating all revisions required by comments from the design development phase.

B. ARCHITECTURAL: Submit or show the following as applicable for the project:

Architectural:	Schematics*	DD* *	CD** *
Location of:			
• Rooms ¹	ü	ü	ü
• Doors ²	ü	ü	ü
• Corridor(s) ³	ü	ü	ü
• Basic column grid/sizes	ü	ü	ü
• Expansion and seismic joints	ü	ü	ü
• Electrical closets	ü	ü	ü
• Equipment rooms	ü	ü	ü
• Signal and telephone closets	ü	ü	ü
• Mechanical shafts and space	ü	ü	ü
• Stair(s)		ü	ü
• Ramp(s)		ü	ü
• Elevator(s)	ü	ü	ü
• Automatic Conveyances	ü	ü	ü
Floor Plans/Drawings:			
• All floors (new and renovated)	ü	ü	ü
• Penthouse	ü	ü	ü
• Roof plan	ü	ü	ü
• Pipe basement	ü	ü	ü
• Pipe tunnel		ü	ü
• Reflected ceiling ⁴		ü	ü
• Equipment floor plans 1:50 (1/4 inch) scale ⁵		ü	ü
• Demolition plans ⁶		ü	ü
Room names and numbers ⁷		ü	ü
Program net/designed net	ü	ü	ü
Exterior dimensions/total building gross area	ü	ü	ü
Size and shape of all departmental functions and services ⁹	ü	ü	ü
Exterior building elevations ¹⁰	ü	ü	ü
Finish floor elevations ¹¹	ü	ü	ü
Door locations, sizes, and swings		ü	ü
Wall thickness and chase walls		ü	ü
Handrail location/dimensions		ü	ü

Architectural:	Schematics*	DD* *	CD** *
Fixed equipment		ü	ü
Equipment elevations and details			ü
Plumbing fixtures		ü	ü
Wheelchair accessible facilities		ü	ü
Wall sections ¹²		ü	ü
Building sections ¹³		ü	ü
Finish grades at corners, entrances, exits, platforms and ramps		ü	ü
Fire and smoke rated partitions ¹⁴	ü	ü	ü
Lead-lined and radio-frequency-shielded partitions ¹⁴		ü	ü
Fire extinguisher cabinets ¹⁴		ü	ü
Spray-on fire proofing (see fire protection)			
Construction details ¹⁵		ü	ü
Drafting symbols, abbreviations, and general notes		ü	ü
Door, window, and louver schedules		ü	ü
Interior details, elevations, sections		ü	ü
Finish schedule ¹⁶		ü	ü
Graphics and signage ¹⁷		ü	ü
Color rendering			ü
Specifications		ü	ü
Lead abatement ¹⁸	ü	ü	
Lead abatement specification ¹⁹		ü	ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

* Submit, as a minimum, a single line layout for at a scale not less than 1:100 (1/8 inch). A scale of 1:200 (1/16 inch) is acceptable for architectural floor layout if an entire floor cannot be shown on one sheet. Submit a complete double line layout of areas of critical importance, at a scale of 1:50 (1/4 inch) including equipment.

** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

B. NOTES:

1. Use lines between spaces to indicate the centerline of the partition (for schematics only).
2. Indicate doors with a slash mark.

3. Along the corridor, the line shall represent the corridor side of the partition.
4. Indicate ceiling mounted equipment, lighting fixtures, air diffusers, registers, tracks, and other significant elements.
5. Identify all equipment for each room. Indicate and coordinate all equipment with the Equipment Guide List (Program Guide 7610) and Activated Equipment List. Use VA standard symbols and notation to distinguish between contractor-furnished and installed (CC), VA-furnished contractor-installed (VC), VA-furnished and installed (VV), VA-furnished with construction funds [VC (CF) and VV (CF)], and relocated (R) equipment. Equipment floor plans are not required for the offices, consultation rooms, classrooms, conference rooms, and waiting rooms within the above departments. Draw equipment details which are necessary for major decisions, though complete detailing is not required for this submittal.
6. Indicate existing finish schedule and notes on plan.
7. Label as required for schematic drawings. Coordinate new room numbering with medical center.
8. Use the same names on drawings as those used in the space program. Provide area figures in fractional form, e.g., 400/390. Indicate space provided, but not called for in the space program, as: -/390.
9. Label each service or activity listed in the Project Scope Data of the Design Program and indicate boundaries with a distinctive line. Include the activity code number (see Handbook 7610).
10. If the project requires exterior work, show all facades indicating massing, proposed fenestration and the building relationship to adjacent structures and the finish grade. Show all significant building materials, including their colors, any proposed roof top mechanical equipment, architectural screens, skylights, and stacks on the elevation drawings. If building is designed for future expansion (vertical and/or horizontal), delineate elevations with and without the future expansion. If project is an addition, show elevations of the existing building in sufficient detail to illustrate the relationship between the new and existing in terms of scale, material, and detail.
11. Define the relationship of the finish ground floor to finish grade at major entrances and docks.
12. Indicate construction including fire resistance rating, building materials and systems, and proposed sill and head heights of openings. Indicate both new and renovated areas on form provided by VA.
13. Define building configuration. Draw sections at the same scale as floor plans, normally 1:100 (1/8 inch). If the building abuts an existing structure, indicate in the section how the new floor elevations align with existing.
14. Identify psychiatric areas where special considerations are required to ensure the safety of patients (e.g. hard ceilings, safety glazing, etc.).
15. Indicate new building components and systems, such as window design, roofing system, special entryways, building "skin", and any special architectural elements for the project. Complete detailing of miscellaneous items is not required for this submission.
16. Indicate all building systems, materials, and future expansion, if applicable.

17. Submit a drawing for all which is part of the construction contract.
18. Provide square meters (feet) of lead paint and x-ray shielding to be removed.
19. Format provided in SPECIFICATIONS. If there is no VA master specification, develop contract specification that is in compliance with regulations of the Environmental Protection Agency.

C. FIRE PROTECTION: Submit the following as applicable for the project:

Fire Protection:	Schematics*	DD* *	CD** *
Fire protection Design Analysis (Narrative): ¹			
• Fire and smoke separation	ü	ü	ü
• Fire sprinkler/standpipe system	ü	ü	ü
• Size of fire pumps	ü	ü	ü
• Water supply available/max. demand	ü	ü	ü
• Water flow testing results	ü	ü	ü
• Fire alarm systems ²	ü	ü	ü
Existing to be modernized	ü	ü	ü
Base loop system for interface of new construction	ü	ü	ü
• Kitchen extinguishing systems	ü	ü	ü
• Size of air handling unit	ü	ü	ü
• Exit paths from each zone	ü	ü	ü
• Distances to stairs	ü	ü	ü
• Occupancy of each area	ü	ü	ü
• Exit calculations for each floor	ü	ü	ü
• Smoke control features	ü	ü	ü
Floor Plans/Drawings: ^{3 & 4}			
• Sprinkler zones	ü	ü	ü
• Fire alarm zones	ü	ü	ü

Fire Protection:	Schematics*	DD* *	CD** *
• Smoke zones	ü	ü	ü
• Building water supply	ü	ü	ü
• Interior sprinkler supply lines	ü	ü	ü
• Standpipes	ü	ü	ü
• Fire extinguisher cabinets	ü	ü	ü
• Fireproofing of structural members	ü		
• Sprinkler/standpipe riser supply piping		ü	ü
• Termination of sprinkler main and inspector test drains		ü	ü
• Sprinkler alarm valves		ü	ü
• Waterflow and tamper switches		ü	ü
• Sprinkler system fire department connections		ü	ü
• Sprinkler design hazards per NFPA 13		ü	ü
• Exit signs and emergency lighting		ü	ü
• Occupied areas not protected by automatic sprinklers		ü	ü
Calculations	ü	ü	ü
Estimated capacities for proposed air handling units in cubic meters (cubic feet) per minute		ü	ü
Location of:			
• Fire alarm system		ü	ü
• Annunciator panels		ü	ü
• Pull stations		ü	ü
• Flow switches		ü	ü
• Audio-visual devices		ü	ü
• Smoke detectors		ü	ü
• Duct smoke detectors		ü	ü
• Smoke dampers		ü	ü
• Fire dampers		ü	ü
• Fire alarm risers ⁵		ü	ü
• Exit signs		ü	ü
• Emergency lighting		ü	ü
• Fire sprinklers		ü	ü
• Standpipes		ü	ü
• Fire hydrants		ü	ü
• Fire pumps		ü	ü
• Post indicator valves		ü	ü
• Sectional valves		ü	ü

Fire Protection:	Schematics*	DD* *	CD** *
• Fire extinguisher cabinets		ü	ü
• Electromagnetic door hold open devices		ü	ü
Wall sections indicating fire resistive ratings		ü	ü
Staff sleeping rooms		ü	ü
Excavation plan signage		ü	ü
Door and window schedule with fire rating or fire rated glazing		ü	ü
Zoning of each fire alarm initiating device		ü	ü
Details:			
• Fire pump system (capacity and pressure)		ü	ü
• Elevation and isometric view of fire pump		ü	ü
• Stairwell sign		ü	ü
• Annunciator panel		ü	ü
Interconnection of fire alarm system with:			
• Smoke dampers		ü	ü
• Air handlers		ü	ü
• Elevator controls		ü	ü
• Kitchen fire extinguishing and fire pump system		ü	ü
• HVAC system with smoke duct detectors		ü	ü
Single line riser diagram for fire alarm system		ü	ü
Height/configuration of storage racks and shelving		ü	ü
FP design/plan review by independent Professional FP Engineer			ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü

* Submit, as a minimum, a single line layout for at a scale not less than 1:100 (1/8 inch). Submit a complete double line layout of areas of critical importance, at a scale of 1:50 (1/4 inch) including equipment.

** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

C. NOTES:

1. Indicate NFPA 220 and UBC fire resistive rating of the building, NFPA 101 occupancy type, and fire protection code analysis to access compliance with NFPA 101.
2. Determine type, features, age, reliability, compliance with present day codes, capacity, zoning, supervision, control panel and power supplies, initiating devices and circuits, and auxiliary functions for existing fire alarm system. Indicate manufacturer, model number, voltage, and wiring style of existing alarm systems and devices. Provide recommendations for the proposed fire alarm work.
3. Provide information to meet JCAHO requirements; e.g. location of all fire rated barriers, smoke barriers, exit signs, fire extinguishers, manual pull stations, smoke detectors, and sprinkler flow switches. Show all interim life safety measures such as temporary systems Fire Alarm, Sprinkler, and Smoke.
4. At DD Submission, add room names, room numbers, door locations and swings, smoke and fire rated partitions, sprinkler/standpipe risers to floor plans. Identify psychiatric areas on drawings so areas for institutional type heads are identified. Add location of all valves (post indicator, sectional) and backflow preventer if provided.
5. Show new equipment and/or the necessary changes involved if modification to the existing system is required. Include any recommendations where certain requirements of VA criteria might be waived, in order to allow the existing equipment to be reused.

D. INTERIOR DESIGN: Submit the following as applicable for the project:

Interior Design:	Schematics*	DD* *	CD** *
Written interior design concept ¹	ü		
Illustrate overall design solution ²	ü		
Material and finish samples	ü		
Sketches	ü		
Design solution for interior spaces:			
• Perspectives		ü	ü
• Plans		ü	ü
• Details		ü	ü
• Elevations		ü	ü
• Sections		ü	ü
• Wayfinding		ü	ü
• Floor patterns		ü	ü
• Wall patterns		ü	ü
• Lighting		ü	ü
• Signage		ü	ü
• Handrails		ü	ü
• Bumper guards		ü	ü
Specification section 09050		ü	ü
Finish schedule		ü	ü
Exterior colors and materials		ü	ü
Sample boards for interior and exterior materials, products, and finishes		ü	ü

Edited carpet and wallcovering specifications		ü	ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü
Keyed Finish plans		ü	ü
Interior design details, elevations, and sections		ü	ü

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** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

D. NOTES:

1. Provide a document of data collected in interior design programming. Include collection and analysis of data from the VAMC project coordinator and interior designer. Data includes, but is not limited to the following: existing interior and exterior design and materials, light, safety, patient profile, customer's "vision" or desired image, public vs. private spaces, complete signage package, goals of customer, relationship to existing facilities, future expansion/renovation plans, regional influences, etc.

2. Discuss and illustrate the overall design solution for the primary areas of the project using marked-up floor plans, loose sketches, and material and finish samples. Use broad categories of materials, finishes, color palettes, patterns, textures, and scales. Separately group all major neutral background materials and finishes that will be used and discuss how they will be integrated with all other materials and finishes on the project. Include all primary and secondary corridors, typical patient and toilet rooms, lobbies, atriums, eating spaces, chapels, waiting rooms, and exam rooms. Show the relationship among departments and functions, and between public and private spaces.

E. STRUCTURAL: Submit the following as applicable for the project:

Structural:	Schematics*	DD* *	CD** *
Three alternative structural systems for typical bays ¹	ü		
Supporting calculations ²	ü	ü	ü
Cost estimates for each system ³	ü	ü	ü
Recommend preferred system	ü	ü	ü
Column locations	ü	ü	ü
Shear load resisting elements ⁴	ü	ü	ü
Boring location plan ⁵	ü	ü	ü
Structural plans ⁶	ü	ü	ü
Sections		ü	ü
Details		ü	ü
Size/location of:			

• Columns		ü	ü
• Beams		ü	ü
Lateral load resisting elements		ü	ü
Load bearing walls		ü	ü
Slabs		ü	ü
Foundations		ü	ü
Elevations		ü	ü
Schedules		ü	ü
General notes		ü	ü
Boring logs		ü	ü
Subsurface investigation report	ü	ü	ü
Estimated quantity of rock		ü	ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

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** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

E. NOTES:

1. When only one structural system is possible due to other project requirements, include an explanatory statement and submit only that structural system.
2. Include vertical and lateral load design for CD submission.
3. Include foundation and fireproofing.
4. Indicate existing utilities and structures within, adjacent, or contiguous to the new construction.
5. Upon approval of the subsurface investigation criteria, submit qualifications of at least three consultants being considered for the work together with the proposal of the consultant recommended as most qualified.
6. If there is only a CD submission, provide a Structural Engineering Analysis Submission within six weeks from the notice to proceed including sketches, calculations, and cost estimates of three alternative structural systems for typical bays, boring location plan for subsurface investigation, and consultant qualifications. For vertical expansion projects, analyze existing structure for structural feasibility.

F. PLUMBING: Submit the following as applicable for the project:

Plumbing:	Schematics*	DD* *	CD** *

Design Analysis (Narrative):			
• Existing plumbing systems to be used and necessary modifications	ü	ü	ü
• New plumbing systems	ü	ü	ü
• New or modified water treatment	ü	ü	ü
Floor Plans/Drawings:			
• Room names	ü	ü	ü
• Identify			
Existing plumbing fixtures w/VA numbering system	ü	ü	ü
New plumbing fixtures w/VA numbering system	ü	ü	ü
Existing equipment	ü	ü	ü
New equipment	ü	ü	ü
New medical gas outlets		ü	ü
New laboratory gas outlets		ü	ü
Plumbing piping	ü	ü	ü
• Size of pipe		ü	ü
• Equipment schedule		ü	ü
• Fire & smoke partitions	ü	ü	ü

Plumbing:	Schematics*	DD* *	CD** *
• Demolition plans		ü	ü
• Riser diagrams		ü	ü
• Legend, notes, and details		ü	ü
Location and size of sprinkler riser, standpipes, and fire pumps (see fire protection)		ü	ü
Location of emergency eyewash and shower equipment		ü	ü
Calculations (equipment & piping)		ü	ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü

F. PLUMBING (cont.):

* Submit, as a minimum, a single line layout for at a scale not less than 1:100 (1/8 inch).

** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics phase.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase. Submit a complete double line layout of areas of critical importance, at a scale of 1:50 (1/4 inch).

G. SANITARY: Submit the following as applicable for the project:

Sanitary:	Schematics*	DD* *	CD** *
Design Analysis (Narrative):			
• Existing sanitary systems: underground water, sanitary sewers, storm sewers, & fuel gas with sources, disposal methods, storage pressures, condition, etc.		ü	ü
• New sanitary systems	ü	ü	ü
• Provide water analysis & expected yield if well required	ü	ü	ü
• Circulation study to assess emergency vehicle access	ü	ü	ü
Install test well, if well is required.	ü		
Utility Plans/Drawings showing existing and new sanitary systems:			
• Size of pipes	ü	ü	ü
• Invert elevations of sewers	ü	ü	ü
• Locate/size			
Pumps	ü	ü	ü
Storage facilities	ü	ü	ü

Treatment equipment	ü	ü	ü
Fire hydrants		ü	ü
Sectional and post indicator valves		ü	ü
Backflow preventer		ü	ü
• Areas of new irrigation system	ü		
• New irrigation system			ü
• Profiles of sanitary & storm sewers			ü
• Demolition Plans		ü	ü
• Legend, notes, and details			ü
Point of connection to sprinkler system	ü	ü	ü
Calculations		ü	ü
List of specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü

G. SANITARY (cont.):

* Submit utility drawings at same scale as provided for Site Development drawings.

** Submit utility drawings at same scale as provided for Site Development drawings, incorporating all of the revisions required by comments from the schematics phase.

*** Submit utility drawings at same scale as provided for Site Development drawings, incorporating all of the revisions required by comments from the design development phase. Submit legend, notes, and details at a scale not less than 1:100 (1/8 inch).

H. HVAC: Submit the following as applicable for the project:

HVAC:	Schematics*	DD* *	CD** *
Description of HVAC systems	ü		
Equipment for each functional space	ü		
Life cycle cost analysis ¹	ü		
Tentative location/sizes:			
• Mechanical equipment room	ü		
• Principal vertical shafts	ü		
Block layout of equipment	ü		
Louvers: ²			
• Outside air	ü	ü	ü
• Exhaust air	ü	ü	ü
• Relief air	ü	ü	ü
Engineering calculations ³	ü	ü	ü
Selection of HVAC equipment		ü	ü
Catalog cuts of equipment		ü	ü
Room by room heating and cooling loads		ü	ü
Zone by zone heating & cooling loads		ü	ü
Building block heating & cooling loads		ü	ü
Tabulation of steam consumption		ü	ü
Psychometric chart for air handling unit		ü	ü
Coil entering and leaving conditions		ü	ü
Fan motor heat gains		ü	ü
Consumption of humidification loads		ü	ü
Sound/acoustic analysis		ü	ü
Room-by-room air balance charts ⁴		ü	ü
Chilled water plant: ⁵			
• Quantity and type of chillers		ü	ü
• Capacity in tons of refrigeration		ü	ü
• Electrical equipment		ü	ü
Heating system:			

HVAC:	Schematics*	DD* *	CD** *
• Total heating load		ü	ü
• Domestic hot water load		ü	ü
• Humidification load		ü	ü
• Equipment steam demand		ü	ü
• Zoning of heating system		ü	ü
HVAC floor plan: ⁶			
• Main supply, return and exhaust ductwork		ü	ü
• Volume dampers		ü	ü
• Fire and smoke partitions		ü	ü
• Fire and smoke dampers		ü	ü
• Smoke detectors		ü	ü
• Automatic control dampers		ü	ü
• Air quantities for each room		ü	ü
• Air inlets/outlets		ü	ü
• Rises and drops in ductwork		ü	ü
• Expansion loops		ü	ü
• Anchors		ü	ü
• Vales		ü	ü
• Drip assemblies		ü	ü
• Balancing fittings		ü	ü
Interconnection of HVAC equipment with fire protection equipment (see fire protection)		ü	ü
Plan/section of mechanical equipment rooms		ü	ü
Schematic flow and riser diagrams ⁷		ü	ü
Schematic control diagrams ⁸		ü	ü
HVAC demolition drawings		ü	ü
Phasing plan		ü	ü
Equipment schedule		ü	ü
Seismic bracing		ü	ü
VA symbols and abbreviation		ü	ü
Selection of			
• Pumps			ü
• Fans			ü
Sizing and selection of			
• Expansion tanks			ü
• Steam to hot water convertor			ü
• Heat exchangers			

HVAC:	Schematics*	DD* *	CD** *
Sound analysis			ü
Complete selection data			ü
Outside chilled water and condenser water distribution ⁹			ü
Standard detail drawings			ü
Automatic temperature control drawings ¹⁰			ü
List of Required Specifications	ü		
Specifications		ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

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** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

H. NOTES:

1. Provide specific design recommendations and full back-up data. Include the heating and cooling capacities of each functional area and the block cooling and heating loads for each new and/or existing building.
2. The locations of these louvers must not allow short circuiting of air from emergency generator exhaust or truck waiting and loading dock areas into air intake etc. Consider factors affecting louver location such as visibility, historical considerations, wind direction, nuisance and health hazard odors (from emergency generator or truck exhausts).
3. Include room-by-room, peak zone-by-zone, and building block heating and cooling loads. Provide a tabulation of steam consumption based on data from all sources. Show correlation between each HVAC zone boundary and architectural floor area correlation between the architectural room numbers and abbreviated/coded room numbers used with computer input data sheets.
4. Show supply, return, exhaust, make-up, and transfer quantities with intended pressure relationships, i.e. positive, negative, or zero with respect to adjoining spaces.
5. Provide pertinent data on accessories such as pumps and cooling tower etc. Show the extent of the outside chilled water and condenser water piping. Clearly show how the piping will be laid in tunnels, trenches, or by direct burial.
6. Show ceiling clearances, at locations where ducts cross each other, by providing 1:50 (1/4 inch) scale local sections. Show all ductwork, and piping 150 mm (6 inch) and larger in double line. Show separate floor plans for air distribution and piping unless waived by VA. Show clearances required for access and maintenance with coil and tube pull.

7. Show typical air handling systems and all hydronic systems with existing capacities and new estimated loads. Verify actual operating conditions and capacities of HVAC systems prior to design.

8. Show control devices, such as, thermostats, humidistats, flow control valves, dampers, freezestats, operating and high limit sensors for all air systems and fluids, smoke dampers, duct detectors etc. Provide a written description of the sequence of operation on the floor plans. Detail the scope of work involved with the Central Engineering Center (ECC) and address if enough spare capacity is available or a new ECC is required. Show a point schedule for analog/digital input/output to be included in ECC.

9. Show pipe sizes and insulation with plans, profile, sections, details, and all accessories, such as, anchors, expansion loops/joints, valves, manholes, capped and flanged connections, interface between the new and existing work (if any). Clearly indicate interferences (if any) with the existing utilities and/or landscape elements on outside piping layout drawings. Show rerouting any utilities, cuttings of roads, pavements, trees, etc., and the extent of new and demolition work. Outside utility drawings shall be based on the study of the latest site drawings, discussions with engineering personnel, and actual site inspection of the existing utility.

10. Show all duct detectors, control valves/dampers static pressure sensors, differential pressure control assemblies, etc., whose actual physical location is critical for the intended sequence of operation on floor plans.

I. ELECTRICAL: Submit the following as applicable for the project:

Electrical:	Schematics*	DD**	CD***
Design Analysis (Narrative):			
• Design ¹	ü	ü	ü
• Life cycle analysis for electrical systems	ü		
Location and size of:			
• Electrical equipment ²	ü	ü	ü
• Electric closets ³	ü	ü	ü
• Telephone closets ³	ü	ü	ü
• Signal closets ³	ü	ü	ü
• Electrical distribution equipment		ü	ü
Drawings showing:			
• Electrical plot plan of existing and proposed underground power (including manholes)	ü	ü	ü
• Telephone systems	ü	ü	ü
• Signal inter-building systems	ü	ü	ü
• Proposed electrical system ⁴	ü	ü	ü
• Electric symbols	ü	ü	ü
• Lighting fixture schedule	ü	ü	ü
• Emergency Life Safety Equipment (see fire protection)		ü	ü
• Symbols, note, abbreviations		ü	ü
List of specialty areas	ü		
Method of short-circuit calculations	ü		

Method of voltage drop and demand calculations	ü		
Utility company correspondence	ü		
Utility company requirements		ü	ü
Load calculations for normal & emergency use	ü	ü	ü
Drawings:			
• Lighting layouts		ü	ü
• Power layouts		ü	ü
• Signal layouts		ü	ü
• Specialty area layouts		ü	ü
• Demolition plans		ü	ü
Riser diagrams		ü	ü
Branch circuit wiring (typ.)		ü	ü
Location and size of:			
• Primary distribution			

Electrical:	Schematics*	DD* *	CD** *
switchgear/switchboard		ü	ü
• Engine-generator sets		ü	ü
• Substation/padmounted transformer		ü	ü
• Manholes		ü	ü
Location of smoke dampers and duct smoke detectors		ü	ü
Interconnection of electrical control equipment with HVAC equipment (see fire protection)		ü	ü
Smoke partitions and fire alarm zones	ü	ü	ü
Fire alarm and signal riser diagrams (see fire protection)		ü	ü
Calculations for emergency generator(s)		ü	ü
Phasing scheme		ü	ü
Electrical details		ü	ü
List of Required Specifications	ü		
Component catalog cuts	ü	ü	ü
Specifications	ü	ü	ü

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** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

I. NOTES:

1. Include basic assumptions, points of interconnection, impact of new construction to existing electrical distribution system, current demand loading (high voltage switchgear and primary feeder), and projected load of new construction. Propose various feasible electrical systems for project and provide advantages/disadvantages.
2. Include means and clearances for installation, maintenance, and removal/replacement of equipment.
3. Electrical, signal and telephone closets must stack vertically.
4. Include high voltage and low voltage switchgear, transformers and low voltage main and/or distribution panels, branch panels and methods of feeding 277/480 volt and 120/208 volt normal and emergency panels.

J. EQUIPMENT: Submit the following as applicable for the project:

Equipment:	Schematics*	DD*	CD**
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		*	*
Equipment (on architectural drawing)	ü	ü	ü
Activation Equipment List (Excel format)		ü	ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

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K. STEAM GENERATION: Submit the following as applicable for the project:

Steam Generation:	Schematics*	DD* *	CD** *
Report on new and existing steam loads ¹	ü		
Life-cycle cost analysis of steam supply alternatives	ü		
Analysis of alternate plant locations	ü		
Life-cycle cost analysis for alternative types of equipment	ü		
Life-cycle cost analysis for heat recovery alternatives	ü		
Data on emissions regulations	ü		
Data on methods of compliance	ü		
Selection of major equipment	ü	ü	ü
Plot plan with new and existing plant locations	ü	ü	ü
Fuel related storage and handling facilities	ü	ü	ü
Alternate plan view layouts of new and existing plant	ü		
Plot plan of steam generating facility ²		ü	ü
Catalog cuts on equipment from two manufacturers		ü	ü
Plans/sections/locations of:			
• Equipment		ü	ü
• Major piping		ü	ü
• Pipe supports		ü	ü
Demolition		ü	ü

Schematic flow diagrams of all piping systems		ü	ü
Calculations:			
• Equipment sizing	ü	ü	ü
• Major piping systems		ü	ü
• Steam load		ü	ü
• Control and regulating valve		ü	ü
• Flowmeter systems		ü	ü
• Steam trap		ü	ü
• Heating and ventilating system		ü	ü
• Steam piping		ü	ü
Schedules		ü	ü
Equipment lists		ü	ü
Verification of emission regulations		ü	ü
List of standards and details		ü	

Steam Generation:	Schematics*	DD* *	CD** *
List of Required Specifications	ü		
Specifications		ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

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** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

K. NOTES:

1. Include maximum and minimum summer and winter demands and total annual production. Provide break-down of new steam loads into categories of end use such as building heating, humidification, reheat, domestic hot water, sterilization, line losses, kitchen, and laundry.

2. Show boilers, pumps, heat recovery devices, tanks, and emission control devices.

L. STEAM DISTRIBUTION (OUTSIDE): Submit the following as applicable for the project:

Steam Distribution (Outside):	Schematics*	DD* *	CD** *
Estimate steam and condensate loads	ü	ü	ü
Life-cycle cost analysis of steam distribution system	ü		
Calculations of pipe sizing	ü	ü	ü
Steam distribution plot plan	ü	ü	ü
Existing underground utilities			
Soil conditions report	ü	ü	ü
Performance requirements for steam traps		ü	ü
Calculate pipe stress		ü	ü
Select expansion facilities for piping		ü	ü
Location of:			
• Manholes		ü	ü
• Pipe expansion devices		ü	ü
Profile drawings including existing utilities		ü	ü
Plan views/sections/dimensions for major piping, pipe layout and pipe supports of:			
• Manholes		ü	ü

• Trenches		ü	ü
• Tunnels		ü	ü
Demolition Plans		ü	ü
List of Required Specifications	ü		
Specifications		ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

* Submit outside steam generation plans at an appropriate scale to show all work involved.

** Submit outside steam generation plans at same scale as topographic/utility survey incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated outside steam generation plans incorporating all revisions required by comments from the design development phase.

M. SOLID WASTE DISPOSAL SYSTEM INCLUDING INCINERATION: Submit the following as applicable for the project:

Solid Waste Disposal System Including Incineration:	Schematics*	DD* *	CD** *
Incineration report including: 1. amount and type of waste (new & existing) 2. emissions regulations and types of emissions controls required 3. life-cycle cost analysis on alternatives for waste disposal 4. calculations of equipment sizing and description of types of equipment 5. viable alternatives for waste disposal	ü		
Evaluation of capability of existing incinerator	ü		
Complete description of existing processing system	ü		
Tests to determine remaining service life and capacity of system	ü		
Plot plan with new plant location and location of existing plant	ü		
Plan view layout of new system or existing system showing new equipment location	ü		
Load calculations on amount and types of waste		ü	ü
Plot plan with location of new processing system		ü	ü
Plans/sections showing locations of:			
• Equipment			
• Major piping		ü	ü
Demolition		ü	ü

Catalog cuts (2 min.) of equipment selections		ü	ü
Emissions control devices		ü	ü
Schedules		ü	ü
Equipment lists		ü	ü
List of standards to be furnished later		ü	ü
List of special details to be furnished later		ü	ü
Verification of applicable emissions regulations affecting design or operation			ü

Solid Waste Disposal System Including Incineration:	Schematics*	DD* *	CD** *
List of Required Specifications	ü		
Component catalog cuts	ü	ü	ü
Specifications		ü	ü
Design Analysis (Narrative)	ü	ü	ü

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** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

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N. AUTOMATIC TRANSPORT: Submit the following as applicable for the project:

Automatic Transport:	Schematics*	DD*	CD*
Automatic transport systems (ATS):			
• Narrative w/ recommended improvements for exiting system	ü	ü	ü
• Traffic study including existing and proposed ATS w/ alternate methods of distribution	ü	ü	ü
Changes to existing systems (arch. dwgs.)		ü	ü
Hoistway (arch. dwg.)		ü	ü
Machine room vents (arch. dwg.)		ü	ü
Type of ventilation (mech. dwg.)		ü	ü
Electrical requirements (elect. dwg.)		ü	ü
Drawings: ^{1, 2, & 3}			
• Automatic Transport Systems		ü	ü
• Elevators		ü	ü
• Dumbwaiters		ü	ü
• Other ATS systems		ü	ü
Sizes/dimensions/details:			
• Hoistway enclosures		ü	ü
• Pits		ü	ü
• Pit ladders		ü	ü
• Machine area ladder and railings		ü	ü
• Entrances		ü	ü
• Machine rooms		ü	ü
Locations/dimensions:			
• Elevator cars		ü	ü
• Entrances		ü	ü
• Counterweights		ü	ü

• Trap doors		ü	ü
Location of hoistway vents		ü	ü
Location of steel hoisting beams		ü	ü
Size of machine beams		ü	ü
Size of end reactions		ü	ü
Location/detail of machine beam pockets		ü	ü
Rail loadings		ü	ü
Hydraulic elevator piston pit loads		ü	ü
Details			
• Hoistway entrances for elevators		ü	ü

Automatic Transport:	Schematics*	DD*	CD*
• Carlifts		ü	ü
• Dumbwaiters		ü	ü
• Trash chutes		ü	ü
• Linen chutes		ü	ü
• ETVS		ü	ü
Elevator machine room equipment layout		ü	ü
Interface with automatic recall and shutdown (see fire protection)			ü
List of Required Specifications	ü		
Specifications		ü	ü
Component catalog cuts	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

* Submit, as a minimum, a single line layout for at a scale not less than 1:100 (1/8 inch). Submit a complete double line layout of areas of critical importance, at a scale of 1:50 (1/4 inch) including equipment.

** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

N. NOTES:

1. Include tracking, piping, battery charging areas, blower rooms, queuing areas, cart holding areas, cart washer, central control area, and floor or wall recessed transport control units. Indicate architectural features in areas to be utilized for these systems. Indicate on architectural drawings all the major equipment located in machine rooms, secondary levels, pits, and the areas pertaining to ATS, AGVS and ETVS.
2. Indicate changes required on the architectural drawings where existing transport systems are retained and modified to serve new and existing areas.
3. Provide all electrical criteria (per basic electrical notes and Automatic Transport Design Manual) on electrical drawings.

O. ASBESTOS ABATEMENT: Submit the following as applicable for the project:

Asbestos Abatement:	Schematics*	DD* *	CD***
Asbestos abatement report including: 1. Summary results of building records 2. Summary results of station personnel interview 3. determination of materials known to contain asbestos 4. visual inspection of building to determine location and condition of asbestos 5. sample strategy on the extent of asbestos present	ü		
Name and location of qualified laboratory for sample analysis	ü	ü	ü
Asbestos abatement drawing		ü	ü
Major Decontamination Areas showing: 1. Limits of sealing off the location 2. Quantities of asbestos material 3. Arrangements for auxiliary rooms 4. Engineering of negative air systems 5. Path of asbestos to loading platform 6. Location and connection to required utilities		ü	ü
Minor Decontamination Areas showing: 1. location, type, and length of pipe element to be abated by "Glove and Bag" approach 2. Other abatement features		ü	ü
Summary of ¹			
• Square meter (feet) of floor space for abatement		ü	ü
• Total linear and square meter (feet) of asbestos to be abated		ü	ü
• Total cost of abatement ²	ü	ü	ü
Asbestos abatement drawings including: 1. restoration of impacted building sub-systems 2. integrated phasing on execution of abatement			ü
List of Required Specifications	ü		
Specifications	ü	ü	ü
Design Analysis (Narrative)	ü	ü	ü

* Submit, as a minimum, a single line layout for at a scale not less than 1:100 (1/8 inch. Submit a complete double line layout of areas of critical importance, at a scale of 1:50 (1/4 inch) including equipment.

** Submit minimum 1:100 (1/8 inch) scale floor plans, new and renovated, incorporating all of the revisions required by comments from schematics.

*** Submit fully dimensioned, complete, and coordinated 1:100 (1/8 inch) scale floor plans, incorporating all revisions required by comments from the design development phase.

O. NOTES:

1. Provide a copy of the summary to the construction cost estimator for inclusion as a separate bid item in the project estimate.
2. Include any cost for decontamination of equipment and fixtures.

P. SPACE PLANNING

	Schematics	DD	CD
Space-Accounting Summary Table	ü ¹	ü ²	ü ³

P. NOTES:

1. Provide a tabular table with columns entitled Departmental Function, H-7610 Requirements, Approved Space Program [Net Square Meters (Net Square Feet)], Variance Between H-7610 and Approved Space Program, Departmental Conversion Factor, Planned Departmental Gross Square Meters (Feet); column totals; and a Total Project Net to Gross Factor. Also, list separately the area required for additions to the program, unassigned space, major circulation (inter-departmental corridors, stairs, elevators), major mechanical and electrical spaces, exterior walls, connecting corridors to other buildings, space for future mechanical system expansion, and similar special requirements.
2. Update table. Justify in writing substantial deviations from the approved space program.
3. Update table.

Q. CRITICAL PATH METHOD (CPM): Submit the following:

Critical Path Method (CPM):	Schematics	DD	CD
Phasing Narrative	ü	ü	ü
Phasing Plans (on reduced site plans)	ü		

Critical Path Method (CPM)j:	Schematics	DD	CD
Phasing Diagram	ü		
Phases (marked on full size drawing)	ü		
Written list of systems ¹	ü	ü	ü
Phasing Diagram (drawn on Phasing Plan) ¹		ü	ü
CPM Phasing Plans (full size contract drawings) ²		ü	ü

Q. NOTES:

1. Include temporary system by phase and separate by technical discipline.
2. One drawing may reflect several reduced site plans.

R. ESTIMATING:

Submit the following:

Estimating:	Schematics	DD	CD
Cost estimate in compliance with Manual for Preparation of Estimates (separate estimates for new construction and alteration work)	ü	ü	ü
Level "A" Summary Sheets for building	ü	ü	ü
Level "A" Summary Sheets for sitework	ü	ü	ü
Building gross area computation (new)	ü	ü	ü
Building gross area computation (alteration work)	ü	ü	
Project Data Sheet 1	ü		ü
Project Data Sheet 1 and 2		ü	ü
Asbestos abatement	ü	ü	ü
Detailed estimate take-off sheets	ü	ü	ü
Level "B" Summary Sheets for buildings		ü	ü
Level "B" Summary Sheets for sitework		ü	ü
Supplement A to SF 252			ü
Detail Market Analysis			ü

S. SPECIFICATIONS

	Schematics	DD	CD
Specifications (All Disciplines)		✓ 1, 2, & 3	✓ 4 & 5

1. Comply with the requirements of the VA Manual for Preparation and Issuance of Construction Solicitation and Contract Documents.

2. Submit for all technical disciplines the original VA Master Specification section drafts marked-up with pencil showing the editing for the project. Clearly identify modifications, deletions and insertions. Assure the specification drafts have been edited and tailored in their application to represent accurate coordination between drawings and specifications.

3. When no VA Master Construction Specification exists for a "unit of work", prepare the specification section consistent with VA Master Construction Specifications format.

a. Use generic or non-proprietary specifications describing the minimal acceptable product criteria level where no "Standard" exists to define quality and workmanship levels.

b. Use applicable "Standards" to define quality and workmanship when these publications exist. List complete designation and title of each publication used in Part 1; follow format in VA Master Construction Specifications for Applicable Publications.

Publications

c. Do not use proprietary specifications or systems that restrict competition unless authorization in writing has been received from the VA Project Manager for such proprietary specification. See the Federal Acquisition Regulation (FAR) Part 10, Part 14, and Part 36.

d. Do not use trade names or manufacturers brand names, except as previously noted.

e. When a deviation is requested, define and specify the minimum acceptable levels of essential criteria in descriptive, physical, functional, or performance requirements.

4. Type specifications in final format and content including any desk copy changes made by the VAMC staff at the previous review. Submit a complete set of the typed specifications for review. Include one set of full-size final drawings of all disciplines, fully coordinated.

5. Return all draft specifications reviewed at DD review to aid the final bid document review. These draft specifications will later be returned to the A/E.

T. FINAL BID DOCUMENTS

- a. Place the seal of the Registered Architect, Registered Landscape Architect, and Professional Engineer responsible for the design and the VAMC Project Director's signature on the Construction Documents. A stamp of the VAMC Project Director's signature will be furnished.
- b. Submit updated Department Ratio Chart of Final Bid stage to the VAMC Project Manager.

III. DISTRIBUTION OF A/E MATERIAL

A. SYMBOL IDENTIFICATION OF CONTRACT DRAWINGS

- AS** - Architectural Drawings (Numbered Only)
HA - Asbestos Removal Drawings
BI - Boring Log Drawings
ES - Electrical Drawings
FA - Fire Protection Drawings
MH - Heating, Ventilating, and Air Conditioning Drawings
PL - Plumbing Drawings
GS - Site Development and Environmental Drawings
CU - Sanitary
and Irrigation
Drawings **MU** -
Steam
Distribution
Drawings **MP** -
Steam
Generation Drawings
SS - Structural Drawings

B. GENERAL NOTES

1. Bond prints shall be full-sized.
2. Bind all drawings into sets in the order of their above classification symbol.
3. All submitted specifications shall be original, unbound, and marked-up VA Master Specifications. Where no VA Master Specification is available, submit a developed specification.

4. Submit all materials, packaged and clearly marked by discipline, to the VA's Contracting Officer. However, where a small amount of material is submitted, the drawings may be packaged together for all disciplines as long as the drawings are separated and tagged with the discipline name. Other material may also be consolidated provided they are labeled and can easily be identified and separated.
5. Material provided unbound will be returned to the A/E. All resubmission costs will be the responsibility of the A/E.

Distribution of A/E Material

Schematic Submission:

VA Medical Center (VAMC)	Appropriate Network Office*
See Supplement "B" A/E Submissions of Design Review Material	0 complete set

Design Development Submission:

VA Medical Center (VAMC)	Appropriate Network Office*
See Supplement "B" A/E Submissions of Design Review Material	0 complete set

Construction Documents Submission:

VA Medical Center (VAMC)	Appropriate Network Office*
See Supplement "B" A/E Submissions of Design Review Material	0 complete set

*Network Office will coordinate the necessary review with the responsible safety and fire protection person in their network.